PSTAT 126: REGRESSION ANALYSIS

FINAL PROJECT

PART - 1
Data Description and Descriptive Statistics

The first objective is to understand the variables in your dataset and their relationships. Your task is to choose a dataset Diamonds from Kaggle: Here's the link. You should include atleast 2 categorical variables and 3 independent quantities.

1. Select a random sample. 2 points
2. Describe all the variables (call summary function on the dataset, see the structure, create histograms for continuous random variable, comment on their distribution, bar plots for categorical random variable). 10 points
3. Choose 3 quantitative and 2 categorical variables appropriately and determine if there is any correlation between these variables. 2 points
4. Run the multiple linear regression model using all these variables and observe the summary statistics. (DO NOT EXPLAIN HYPOTHESIS TESTING OR ANYTHING ELSE) 2 points
5. Comment on anything of interest that occurred in this part. Were the data approximately what you expected, or did some of the results surprise you?

Part - 2

SIMPLE LINEAR REGRESSION

PART I continuation...

1. Start with one predictor and one response from the variables you chose in Part I. For instance, you can start with the predictor 'carat' and the response 'price', and conduct a simple linear regression analysis on it. 2 points

2. Run the model and examine the summary statistics, interpreting everything (hypothesis testing, R^2_{adj} as discussed in class, confidence interval, prediction interval, plot, etc.). 7 points

3. Test the assumptions and apply any necessary transformations to the response variable y or the predictor.
   7 points

4. Call the summary function on the transformed variables, observe the summary, and note any changes. 2 points

5. Add other variables to the model and assess if the model improves. For step 5, run the code in the background and include all interpretations in the file. For instance, if adding depth to the simple linear regression model (carat and price) increases the adjusted R^2, include it in the model; if it decreases, exclude it. Do not include the code for step 5 in the submitted file; only write the conclusions. 6 points

6. Comment on anything of interest that occurred while doing this part.

Part -3

PART II continuation...

1. Based on the best model obtained from Part II (you would have more than one variable now), run it and call the summary function to analyze how it works and what you observe.

   OR

   In class we saw different techniques and criterion to find best model. You can use any method and technique you prefer (e.g., backward elimination using AIC or stepwise regression using AIC or backward elimination using BIC criterion) to find the best model and document your observations. 10 points

2. Detect multicollinearity among the variables using the variance inflation factor (VIF). 2 points

3. Give CIs for a mean predicted value and the PIs of a future predicted value for at least one combination of X's (from your final linear model). 4 points

4. Summarize your report (for the final deliverable). 4 points